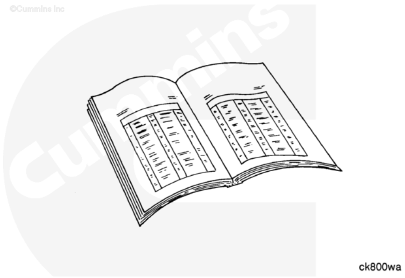
Trainings

Preparatory Steps

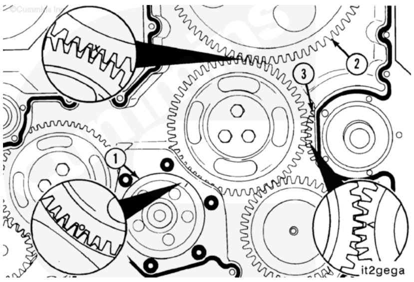
- Remove the gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html)



Remove

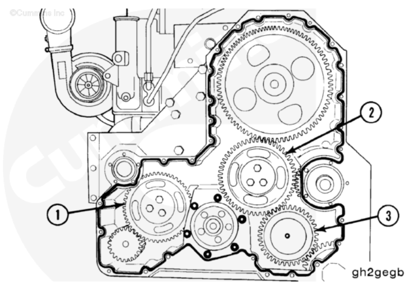
Before removing the idler gear, use the accessory driveshaft to rotate the crankshaft until the timing marks on the crankshaft gear (1), camshaft gear (2), and accessory drive gear (3) are completely engaged in the camshaft idler gear.

Note : Due to the camshaft idler gear having more gear teeth than the crankshaft gear, the camshaft idler gear timing marks will not align with the crankshaft, camshaft, and accessory drive gears timing marks on every engine revolution.



Three idler gear assemblies are used:

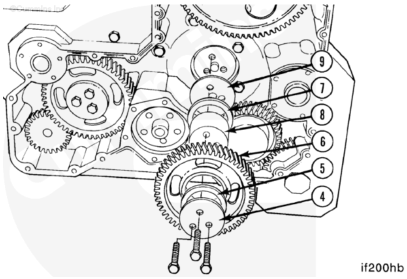
- Water pump idler gear (1)
- Camshaft idler gear (2)
- Hydraulic pump idler gear (3).



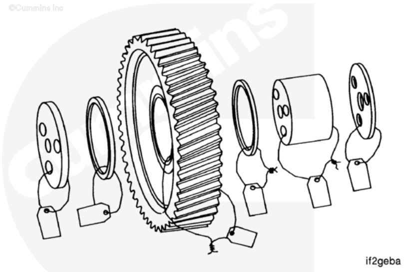
To remove the camshaft idler gear assembly, remove:

- The three capscrews
- Cover plate (4)
- Front thrust bearing (5)

- Idler gear (6)
- Rear thrust bearing (7)
- Idler gear shaft (8)
- Wear plate (9).



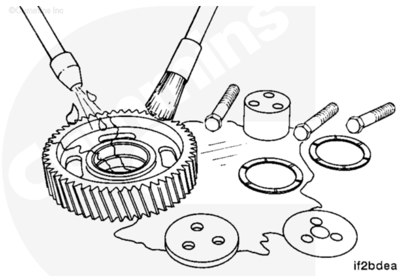
Mark or tag the individual pieces of the idler gear assembly and attach them together.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

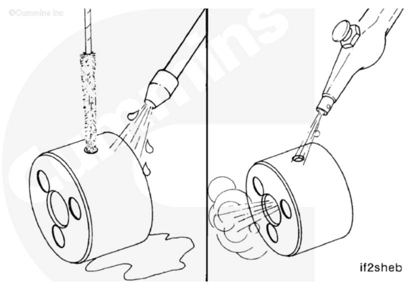
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the parts with solvent.

Dry with compressed air.

⚠ WARNING ⚠

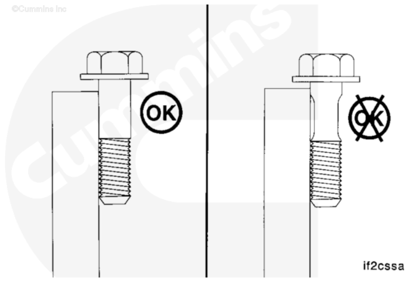
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



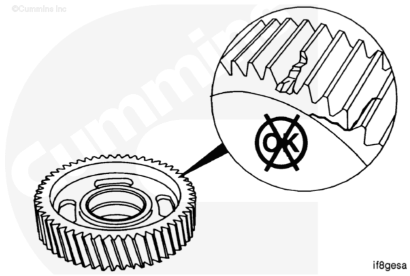
Use a bristle brush to clean the oil drillings in the idler gear shaft.

Blow out the oil drillings with compressed air.

Use a straightedge to inspect the retaining capscrews for necking. If necking is observed, do **not** reuse the capscrews. They **must** be replaced.

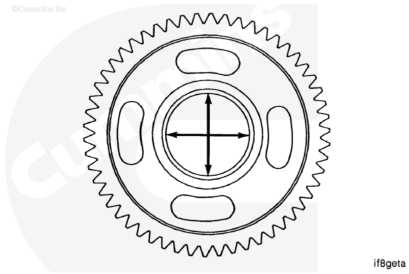


Inspect the idler gear for chipped, broken, or cracked teeth.



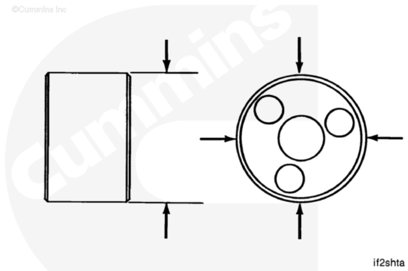
Measure the inside diameter of the idler gear bushing bore.

Camshaft Idler Gear Bushing Bore Inside Diameter		
mm		in
60.045	MIN	2.3640
60.100	MAX	2.3661



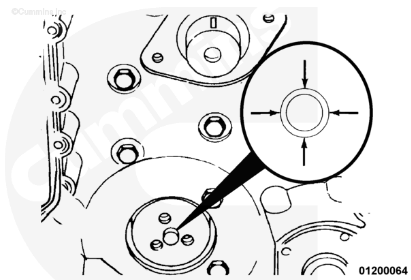
Measure the outside diameter of the idler gear shaft.

Camshaft Idler Gear Shaft Outside Diameter		
mm		in
59.975	MIN	2.3612
60.008	MAX	2.3625



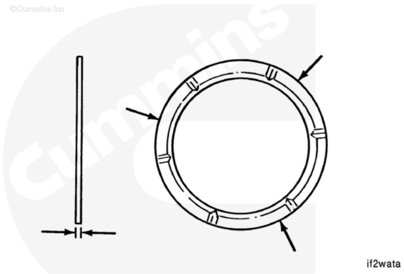
Measure the outside diameter of the camshaft idler shaft ring dowel.

Camshaft Idler Ring Dowel Outside Diameter		
mm		in
19.217	MIN	0.7566



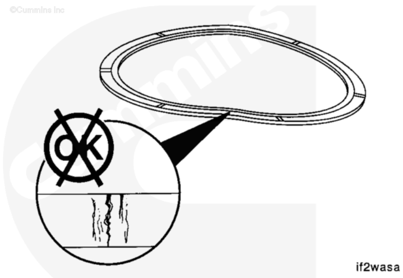
Camshaft Idler Ring Dowel Outside Diameter		
mm		in
19.243	MAX	0.7576

Measure the thrust washer thickness in three places 120-degrees apart.



Camshaft Idler Thrust Washer Thickness		
mm		in
2.400	MIN	0.0945
2.470	MAX	0.0972

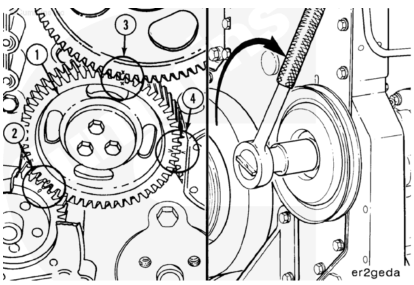
Flex the thrust washers approximately 3 to 6 mm [1/8 to 1/4 in], and inspect the surfaces for cracks.



Install

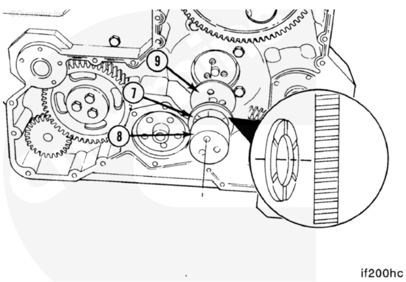
When installing the camshaft idler gear (1), make certain the timing mark “0”on the crankshaft gear (2), timing mark “X” on the camshaft gear (3), and timing mark “V” on the accessory drive gear (4) are aligned as shown.

The marks on the idler gears **must** match the same mark on each of the other gears.



⚠ CAUTION ⚠

The grooved side of the rear thrust bearing must be facing toward the gear to prevent damage to the gear and engine during engine operation.



Note : Only the camshaft idler gear has a wear plate.

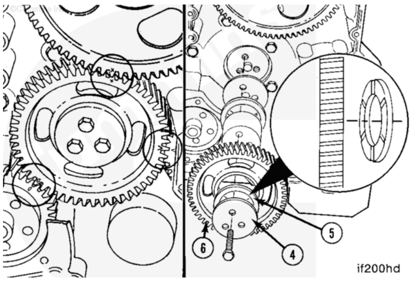
Use Lubriplate™ 105 or equivalent, to lubricate the wear plate, thrust bearing, and idler gear.

Install the camshaft idler gear wear plate (9).

Install the idler gear shaft (8) and rear thrust bearing (7).

⚠ CAUTION ⚠

The grooved side of the front thrust bearing must be facing toward the gear to prevent damage to the gear and engine during engine operation.



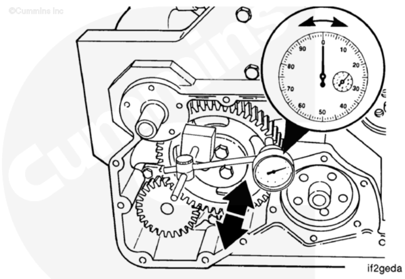
Align the timing marks and install the idler gear (6), front thrust bearing (5), and gear retainer (4).

Install the retaining capscrews and tighten.

Torque Value:

1. 61 n•m [45 ft-lb]
2. Rotate 60 degrees

The following idler gear end clearance and backlash checks apply to all three idler gears. The same MINIMUM and MAXIMUM values apply, also.

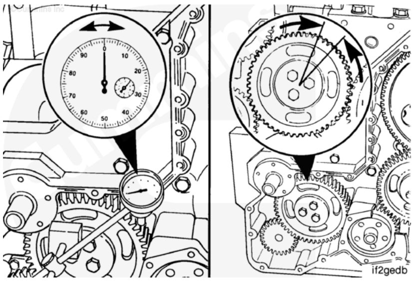


Use a dial indicator gauge with a magnetic base to measure the idler gear end clearance.

Place the contact tip of the gauge against the face of the idler gear.

Idler Gear End Clearance		
mm		in
0.30	MIN	0.012
0.53	MAX	0.021

Use a dial indicator gauge with a magnetic base to measure the idler gear backlash.



Place the contact tip of the gauge against a tooth on the idler gear.

Note : Do not allow the mating gears to move while measuring the backlash.

Idler Gear Backlash		
mm		in
0.08	MIN	0.003
0.38	MAX	0.015

Finishing Steps

- Install the gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html)
- Operate the engine to normal operating temperature and check for leaks.

Last Modified: 21-Jan-2009